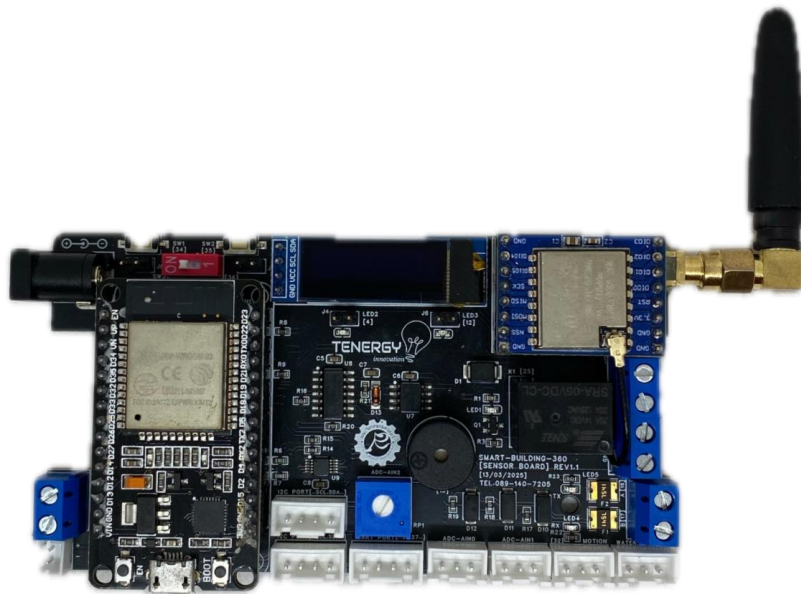


Model: Tenergy32Hub – ESP32-Based
Designed & Manufactured by: TENERGY
INNOVATION CO., LTD.

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Product Overview

The Tenergy32Hub is a robust and versatile ESP32-based IoT gateway board designed for smart buildings, industrial automation, and sensor network applications. This board integrates multiple communication interfaces and sensor connectivity options, making it suitable for a wide range of smart system deployments.

Technical Specifications

- - Microcontroller: ESP32-DOIT-DEVKIT-V1 (dual-core, Wi-Fi + BLE)
- - Power Supply: Supports wide-range DC input from 4.5V to 60V
- - Battery Support: 3.7V Lithium battery input with onboard charging circuit
- - Analog Input: 2-channel ADC (0-5V) with 16-bit resolution

- - Display: Integrated 128x32 OLED display
- - LoRa Module: SLX1278 operating at 433MHz for long-range communication
- - User Interface: 2x Tactile switches and 1x Slide switch for configuration and input
- - Indicator LEDs: Red and Blue LEDs for programmable system status
- - Buzzer: 2.048 kHz onboard piezo buzzer
- - Relay Output: 1x Relay switch with COM and NO terminals
- - RS-485 Port: Industrial-grade communication port compatible with PLCs, Schneider power meters, industrial sensors, HVAC systems, motor drivers, and more
- - I2C Port: For connection with I2C sensors such as SHT30 (temperature/humidity), BH1750 (light), MPU6050 (vibration), BMP280 (pressure), etc.
- - UART Port: For serial communication with NEO-6M/NEO-M8N (GPS modules), MH-Z19B (CO2 sensor), RC522/EM18 (RFID readers), and RS232-compatible devices
- - Networking: Supports mesh networking via ESP-NOW, Wi-Fi mesh, or LoRa for large-area coverage

Specifications Overview

Feature	Description
Microcontroller	ESP32-DOIT-DEVKIT-V1 Dual Core 240MHz, WiFi + BLE
Operating Voltage	DC 4.5V to 60V wide input range
Battery Support	3.7V Lithium Battery Input (JST connector) with onboard charging circuit
Display	OLED Display 128x32 pixels for real-time status or configuration feedback
LoRa Module	SLX1278 (433 MHz) for long-range wireless communication
Analog Inputs	2x ADC (0–5V), 16-bit resolution
Switches	2x Tactile Push Buttons (Tack Switch), 1x Slide Switch for configuration and user interaction
LED Indicators	1x Red LED, 1x Blue LED (programmable)
Buzzer	Built-in 2.048 kHz Piezoelectric Buzzer
Relay Output	1x Relay Module (COM, NO), suitable for switching AC/DC loads
RS-485 Port	Industrial-grade RS-485 interface for Modbus RTU communication with:

- PLCs (Siemens, Omron, Mitsubishi)
- Digital Power Meters (e.g., Schneider PM5000 series)
- HVAC controllers, motor drives
- Industrial sensors and actuators |

| **I2C Interface** | For external sensors like:

- Temperature/Humidity: SHT30, AHT20
- Light: BH1750
- Vibration: MPU6050, ADXL345
- Pressure: BMP280 |

| **UART / RS232 Interface** | Compatible with devices such as:

- GPS Modules: NEO-6M, NEO-M8N
- CO2 Sensor: MH-Z19B
- RFID Reader: RC522, EM18
- Thermal Sensor: MLX90614
- Serial-to-RS232 converters and RS232-based equipment |

| **Battery Monitoring** | Built-in voltage divider circuit for real-time battery level (%) detection |

| **Digital I/O** | 2x configurable GPIO ports for digital input/output control |

| **Mesh Networking** | Supports ESP-NOW, Wi-Fi mesh, and LoRa-based mesh networks for large-area wireless communication |



Applications

- Smart Building Gateway
 - Industrial IoT Automation Node
 - Agricultural Data Logger
 - Remote Monitoring System
 - Educational IoT Platform
-



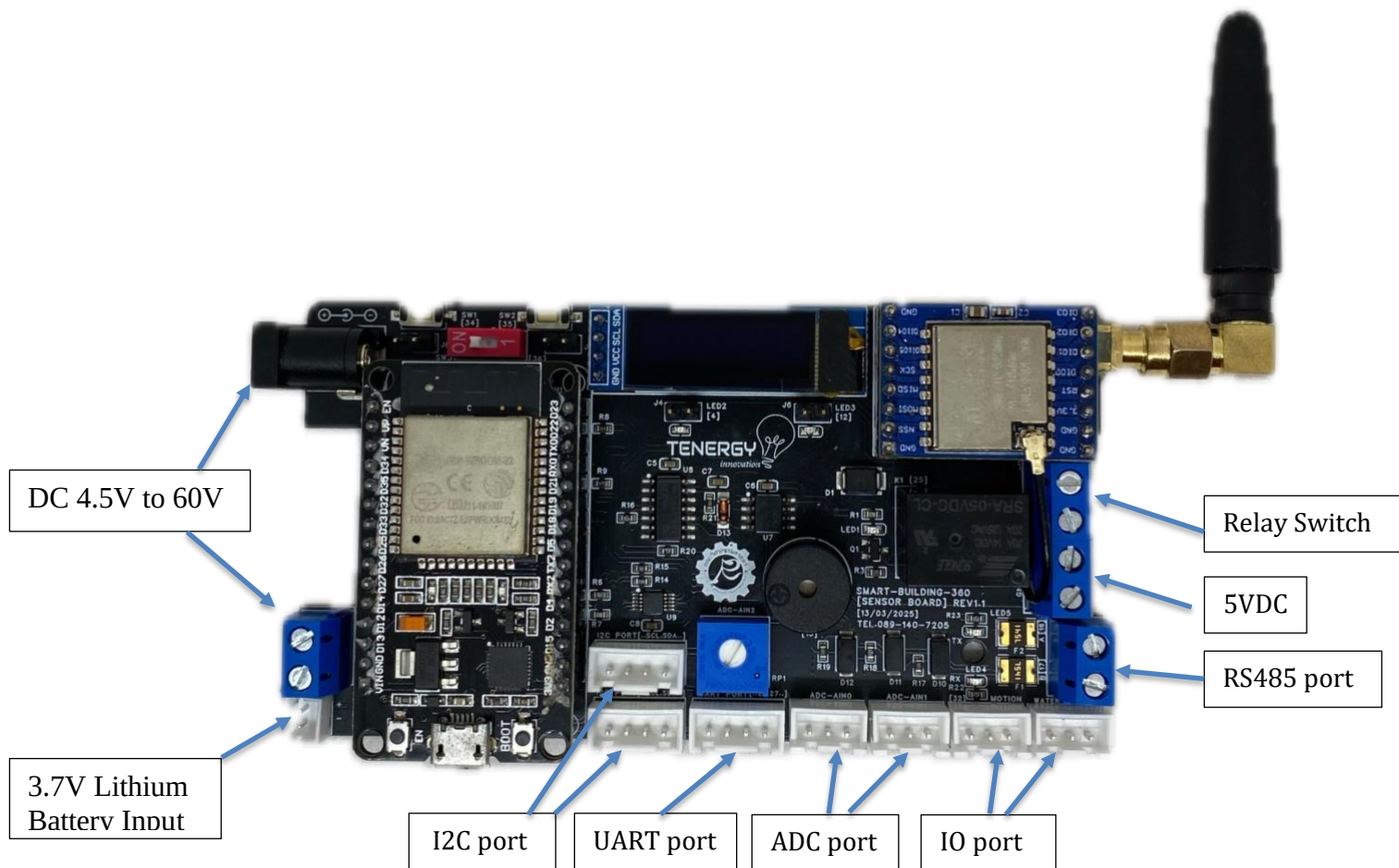
Board Dimensions & Mounting

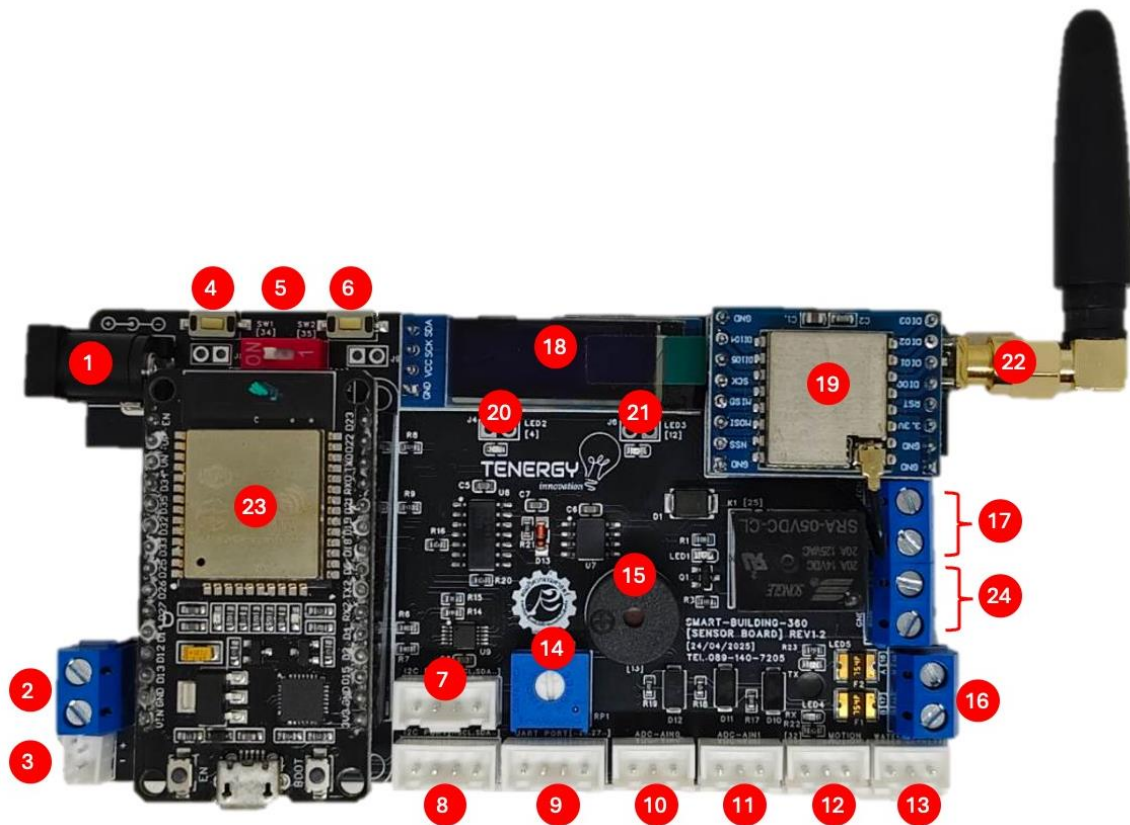
- Size: ~ 120mm x 70mm
- 4x M3 mounting holes
- Antenna: SMA connector with external antenna (LoRa)



Development Support

- **Programming Interface:** USB (via onboard ESP32 module)
 - **Firmware Platform:** Arduino (PlatformIO / Arduino IDE)
 - **OTA Support:** Wi-Fi OTA firmware update supported
 - **Example Libraries:** LoRa, OLED, RS485, ADC, I2C, UART, Mesh Networking
-





หมายเลข	รายละเอียด	เพิ่มเติม
1	DC Power Jack	12 V DC adapter jack to power the board (connects directly to terminal 5V)
2	DC-IN Terminal	12 VDC terminal block for board power
3	Battery Terminal Connector	Connector for a 3.7 V lithium battery
4, 6	Tactile Switches	Push-buttons SW1 & SW2 on ESP32 GPIO 34/35 for programming and testing
5	DIP Switches	Slide switch SW3 on ESP32 GPIO 36 for programming and testing
7-8	I2C Connector	I ² C header (SDA & SCL) for external I ² C devices
9	UART Connector	UART header on ESP32 pins 26/27 (can also serve as digital inputs)
10	ADC0 Connector	0–5 V analog input to AS1115 port AIN0
11	ADC1 Connector	0–5 V analog input to AS1115 port AIN1
12	Motion Connector	Digital input for motion sensor on ESP32 pin 32
13	Water Leak Connector	Digital input for water-leak sensor on ESP32 pin 33

14	Potentiometers	Adjustable resistors on AS1115 port AIN2
15	Buzzer	2.048 kHz piezo buzzer on ESP32 pin 13
17	RS485 Connector	RS-485 interface (UART) for long-distance, noise-immune communication
17	Relay	COM/NO relay switch on ESP32 pin 25
18	OLED Display	0.91" 128 × 32 px I ² C display for status & sensor readings

19	LoRa Module SLX1280	SLX1280 LoRa 433 MHz SPI module for gateway communication
20	Blue LED	Blue indicator LED on ESP32 pin 12
21	Red LED	Red indicator LED on ESP32 pin 4
22	LoRa Antenna	SMA antenna connector for LoRa module
23	ESP32-DOIT-DEVKIT-V1	ESP32 Main microcontroller of the board
24	5VDC-Out Connector	5 VDC output to power external devices
25	Charging Module	On-board charging circuit for a 3.7 V lithium battery